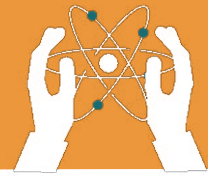


UNIT 5

MATERIALS AND THEIR PROPERTIES

Learning objectives:

- Understand material properties like hardness and flexibility.
- Differentiate metals and non-metals by properties and uses.
- Identify materials and their practical applications.



MATERIALS AND THEIR PROPERTIES

Materials are substances that make up everything around us. From the clothes we wear to the buildings we live in, and even the devices we use daily – all are made from different types of materials. Understanding materials and their properties helps us choose the right material for the right purpose.

Every material has unique properties that make it suitable for specific uses:

5.1 Physical Properties

Materials have different physical properties that help us understand how they behave and where they can be used. Some important physical properties include:

- **Hardness:** The ability of a material to resist scratching or denting, e.g. diamonds are the hardest natural material.

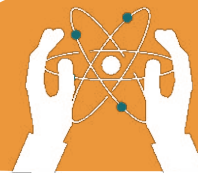


- **Texture:** The surface feel of a material, such as smoothness, roughness, or softness.

- **Flexibility:** How easily a material can bend without breaking. Rubber is very flexible, while glass is not.



MATERIALS AND THEIR PROPERTIES



- **Transparency:** The ability of a material to allow light to pass through it. Glass is transparent, while wood is opaque.



5.2 Metals and Non-Metals

Elements are broadly classified as **metals** and **non-metals** based on their properties:

Property	Metals	Non-Metals
Physical State	They are hard and shiny	They are soft and dull
Malleability	They are malleable (can be hammered into shapes)	They are brittle (break easily)
Conductivity	They are good conductors of heat and electricity	They are poor conductors of heat and electricity
Examples	Iron, Aluminium, Copper	Carbon, Sulfur, Oxygen
Uses	They are used in construction, making wires	They are used in fertilizers and medicines



MATERIALS AND THEIR PROPERTIES

Metals



Gold



Copper



Iron



Nickel



Silver

Non-Metals



Diamond



Oxygen



Carbon



silicon



boron

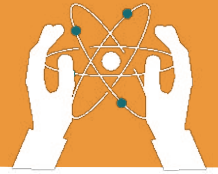
5.3 Everyday Materials and Their Importance

We use various materials in our daily lives, such as:

- **Plastic:** Light, durable, and waterproof; It is used in making bottles and containers.
- **Glass:** Transparent and hard; used in windows and jars.
- **Wood:** Strong and flexible; it can be used in furniture and construction.
- **Fabric:** Soft and versatile; they are used in making clothing and bedding.

Each material has unique properties that make it suitable for specific purposes. Understanding these properties helps us choose the right material for the job.

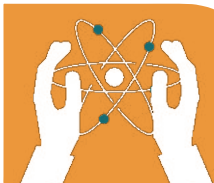
MATERIALS AND THEIR PROPERTIES



EXERCISES

Fill in the blanks.

1. _____ is a material that can bend without breaking.
2. Metals are good conductors of _____ and electricity.
3. _____ is a transparent material used in windows.
4. Wood is _____ and flexible, making it ideal for furniture.
5. _____ is the hardest natural material.
6. Non-metals are usually _____ and brittle.
7. _____ is a material used for making wires due to its conductivity.



MATERIALS AND THEIR PROPERTIES

Tick(✓) for True and cross (x) for False statements

1. Plastic is a good conductor of electricity.
2. Metals are usually soft and brittle.
3. Glass is an example of a transparent material.
4. Non-metals are poor conductors of heat.
5. Wood is heavier than steel.
6. Hardness is a physical property of materials.
7. Copper is a type of non-metal.
8. Fabric is commonly used in clothing.

Choose the correct answer

1. The hardest natural material is:

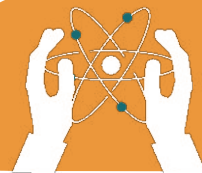
- | | |
|------------|------------|
| a) Wood | b) Plastic |
| c) Diamond | d) Iron |

2. Metals are good conductors of:

- | | |
|--------------------------|----------------------|
| a) Heat and electricity | b) Water and light |
| c) Sound and temperature | d) None of the above |

3. is a flexible material:

- | | |
|-----------|--------------|
| a) Rubber | b) Glass |
| c) Steel | d) Aluminium |



4. Non-metals are typically:

- a) Shiny and malleable
- b) Dull and brittle
- c) Good conductors of electricity
- d) Heavy and shiny

5. is used in making furniture:

- a) Glass
- b) Wood
- c) Plastic
- d) Metal

6. A good conductor of electricity is:

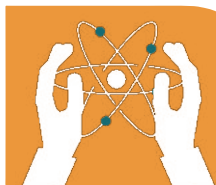
- a) Copper
- b) Sulfur
- c) Plastic
- d) Rubber

7. The term used to describe how much a material resists scratching is:

- a) Transparency
- b) Hardness
- c) Flexibility
- d) Conductivity

8. Which material is used for making windows?

- a) Wood
- b) Glass
- c) Rubber
- d) Plastic



Answer the Following Questions:

Short-Answer Questions

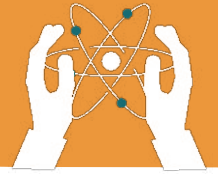
- 1. Name three physical properties of materials.**
- 2. Give two examples of non-metals and metals.**
- 3. What are materials?**

Long-Answer Questions

- 1. Explain the physical properties of materials with examples.**
- 2. Write the difference between metals and non-metals.**
- 3. Describe the uses of everyday materials in daily life.**

Reasoning Questions Answer

- 1. Why is copper used in making wires?**
- 2. Why is it important to understand the material?**



Physical Properties of Matter

Join the pictures from the left to the sentences on the right



Solubility



Magnetic force



Insulators



Hardness



Good conductors



Transparency

MALLEABILITY



Malleability

_____ is the property of a material to not change shape when squeezed.

Materials that stop heat or electricity are called _____ or bad conductors.

The property of one material dissolving inside another is called _____.

The property of being able to see through a material is called _____.

The property of changing the shape of a material without it breaking is called _____.

The ability of a magnet to attract some materials is known as _____.

Gold, metal spoons, wires, and water are examples of _____ of heat and electricity.



MATERIALS AND THEIR PROPERTIES

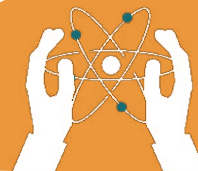
ACTIVITY

- Collect objects made of different materials (e.g., plastic bottle, metal spoon, wooden block, glass jar).
- Test their properties (e.g., hardness, transparency, and conductivity).



Critical Thinking

- Why can't we use glass to make bicycle frames or rubber to build bridges? What properties make these materials unsuitable?
- Why do you think metals are preferred over non-metals for making electrical wires? Explain your reasoning with examples.



Properties of Materials

Activity 1

1.1. What does 'physical property' mean?

1.2. There are a few physical properties. Complete the following table:

Physical property	Description	Example
a) Conductivity		
b) Hardness		
c) Insulation		
d) Transparency		
e) Solubility		